Chapter 8

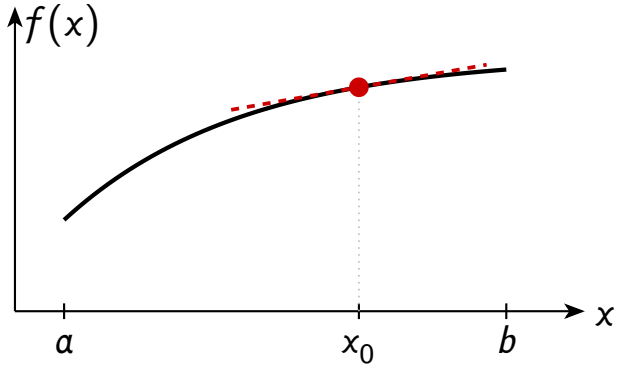
# Numerical differentiation

# Numerical differentiation - goal

**Goal**: given $f(x)$, $x \in [a, b]$, compute the derivative of $f$ at a point $x_0$

Analytically: limit of the incremental ratio

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0} = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$




Geometrically: slope of the tangent to $f$ at $x_0$ (red, dashed)

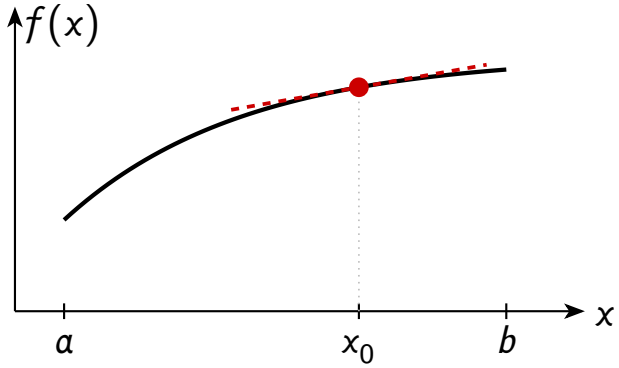
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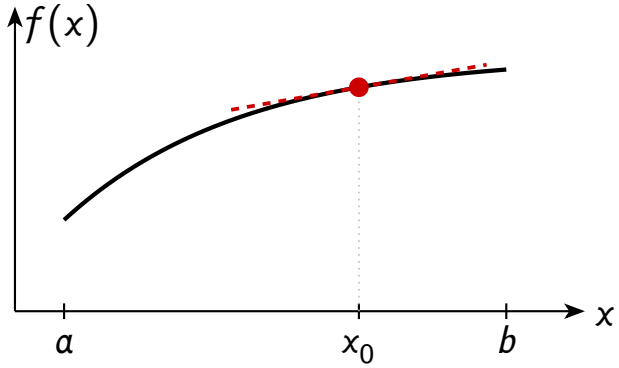
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# Numerical differentiation - nodal values

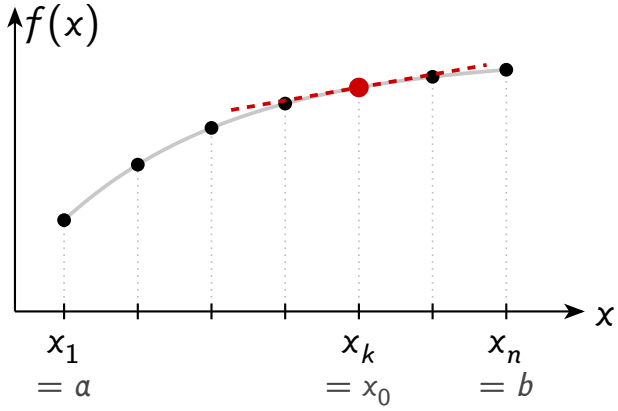
Numerically: $f$ is known only at a **finite** number of points, e.g. the unknown of a field problem computed on a grid

Subdividing $[a, b]$ with $n$ **nodes** ($n-1$ intervals):

$$x_1 = a, \quad x_2, \quad \ldots, \quad x_k = x_0, \quad \ldots, \quad x_n = b$$

**Nodal values**: the values of $f$ at the nodes, $f_k = f(x_k)$, $k = 1, \ldots, n$

gray curve: $f$ between the nodes is not available




**Idea**: estimate the derivative at $x_k$ using only the nodal values $f_1, f_2, \ldots, f_n$

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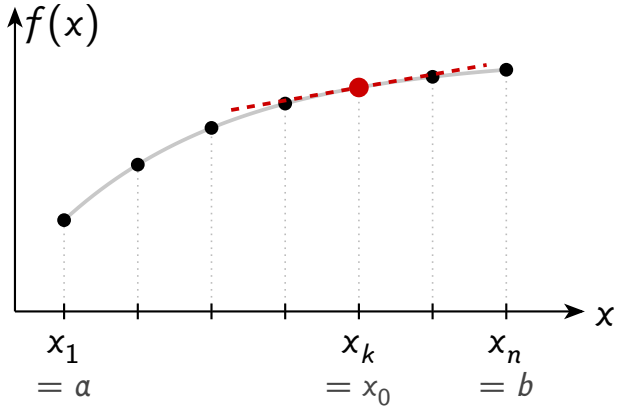
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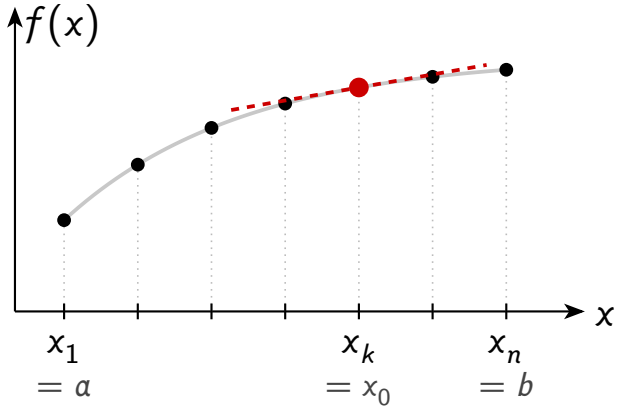
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# Taylor expansion

First derivative

Second derivative

# Numerical differentiation - from nodal values to the derivative

**Question**: the nodal values $f_{k-1}, f_k, f_{k+1}, \ldots$ are just numbers. How do we get the derivative at $x_k$ from them?

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The derivative is not among the nodal values: we need a relation between

- the value of $f$ at a **neighbouring** node, e.g. $f_{k+1}$
- the value **and the derivatives** of $f$ at $x_k$

Such a relation exists: the **Taylor expansion** of $f$ around $x_k$

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# Taylor expansion - series

## Taylor expansion of $f(x)$ around $x_0$

$$f(x) = f(x_0) + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0} (x - x_0) + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{x_0} (x - x_0)^2 + \left.\frac{1}{6}\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right|_{x_0} (x - x_0)^3 + \ldots$$

**Note**: infinitely many terms; coefficient of the power $m$: $1/m!$

Truncating after the second-order term:

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$O\left((x-x_0)^3\right)$ stands for the error $E(x)$ introduced by truncating the Taylor series

$$|E(x)| = \left| f(x) - \left[ f(x_0) + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0} (x-x_0) + \frac{1}{2}\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{x_0} (x-x_0)^2 \right] \right|$$

$E(x) = O\left((x-x_0)^3\right)$ means:

$$\exists M \in \mathbb{R}^+ : \quad |E(x)| \leq M \, |x-x_0|^3 \quad \text{for } (x-x_0) \to 0$$

"there exists a positive real number $M$ such that, for $x$ close to $x_0$, $|E(x)|$ is at most $M \, |x-x_0|^3$"

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# Taylor expansion - halving the distance

**Question**: error $O\left((x-x_0)^N\right)$. What happens if the distance $(x-x_0)$ is halved?

Distance $(x-x_0)$: error $E \approx M\ |x-x_0|^N$; new point $x'$ with $(x'-x_0) = \frac{1}{2}(x-x_0)$:

$$E' \approx M\ |\frac{1}{2}(x-x_0)|^N$$

$$\downarrow\ |1/2\ (x-x_0)|^N = (1/2)^N\ |x-x_0|^N$$

$$= \left(\frac{1}{2}\right)^N M\ |x-x_0|^N \approx \left(\frac{1}{2}\right)^N E$$

Example, $N=2$: $E' \approx (1/2)^2 E = E/4$

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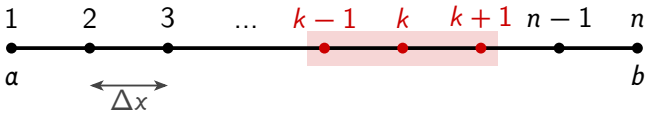
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# Finite differences - Taylor expansion between nodes

**Hp**: uniform spacing, $x_{k+1} - x_k = x_k - x_{k-1} = \Delta x$




Taylor expansion around $x_0 = x_k$, evaluated at the next node $x = x_{k+1}$:

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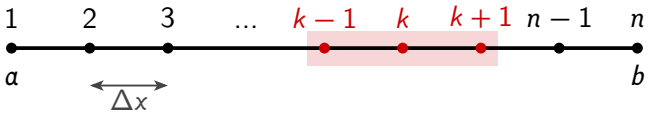
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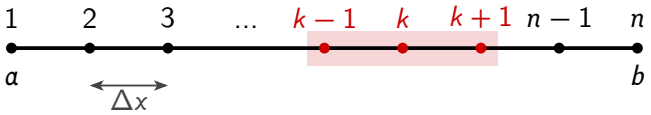
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# Finite differences - forward formula: truncation

Starting from the expansion [1] to the next node

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the second-order term is now part of the error

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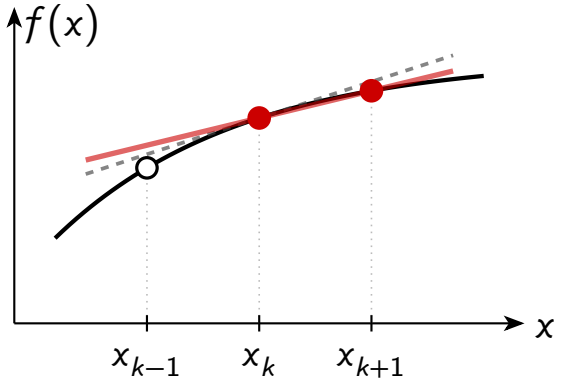
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red: secant, forward formula

dashed: tangent (exact)

Reordering:

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**forward** finite difference: **first-order** accurate

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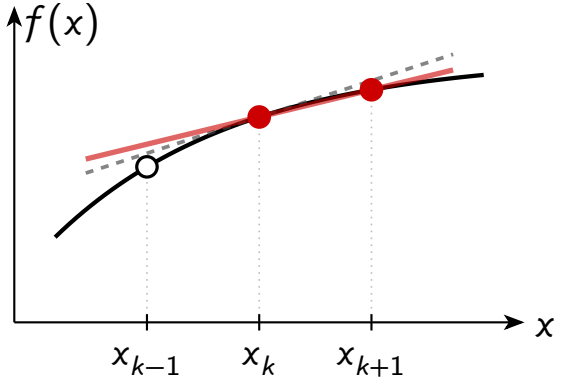
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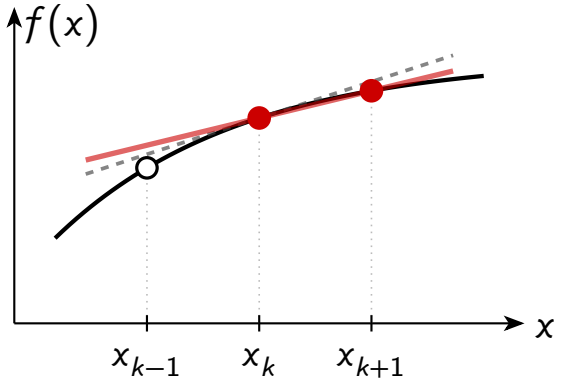
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Reordering:

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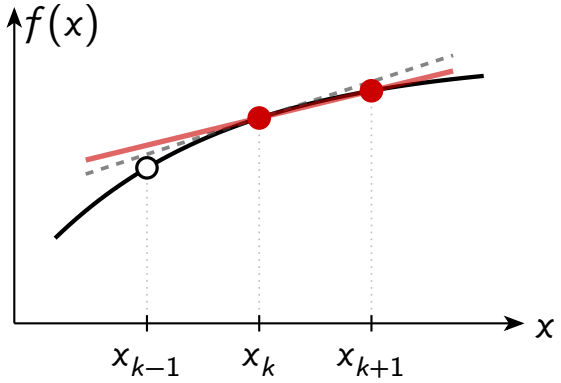
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Same expansion around $x_0 = x_k$, now evaluated at the **previous** node $x = x_{k-1}$

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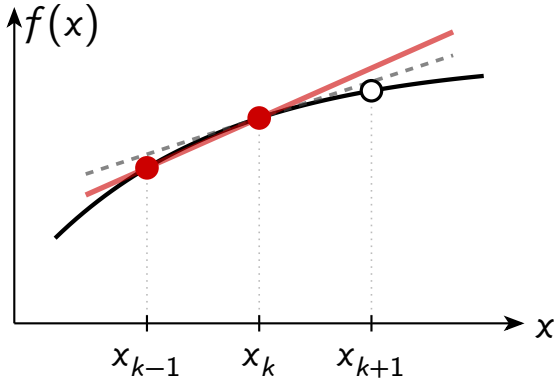
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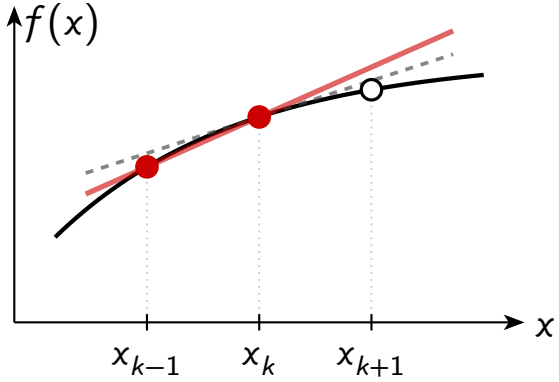
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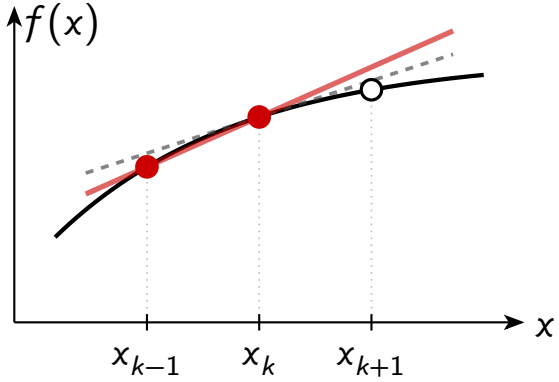
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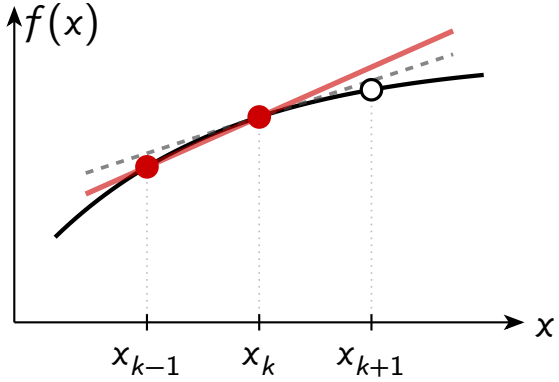
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Origin of the $O(\Delta x)$ error: the second-derivative term of [1] (or [2]), moved into the error and divided by $\Delta x$

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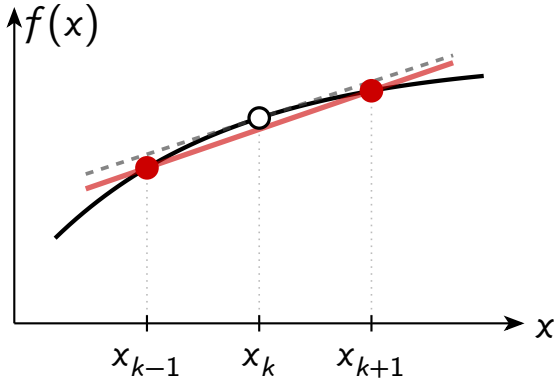
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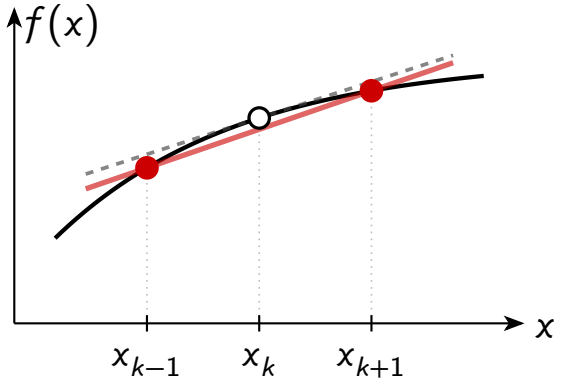
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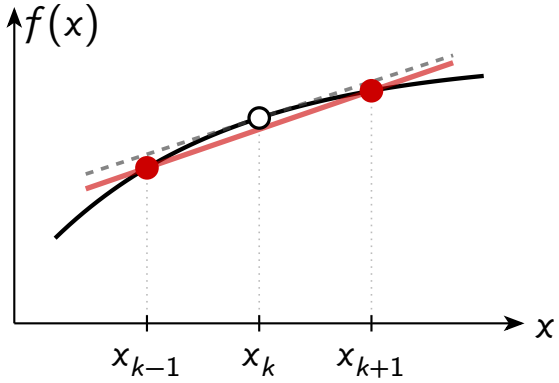
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# Finite differences - centered formula

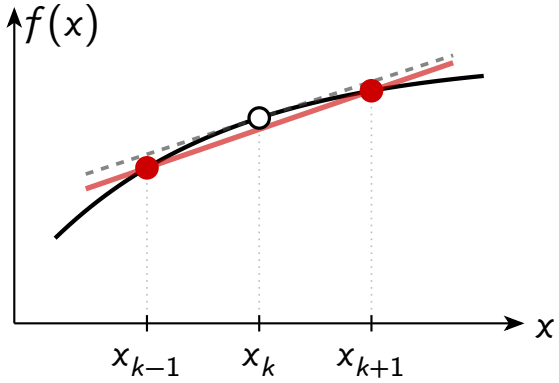
$$f_{k+1} - f_{k-1} = 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + O(\Delta x^3)$$

$$\downarrow \text{ dividing by } 2\Delta x$$

$$\frac{f_{k+1} - f_{k-1}}{2\Delta x} = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k + \frac{O(\Delta x^3)}{2\Delta x}$$

$$\downarrow O(\Delta x^3)/(2\Delta x) = O(\Delta x^2)$$

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red: secant, centered formula

dashed: tangent (exact)

Reordering:

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k = \frac{f_{k+1} - f_{k-1}}{2\Delta x} + O(\Delta x^2)$$

**centered** finite difference: **second-order** accurate

# Finite differences - one-sided second-order formula

**Question**: a **second-order** formula using only nodes on **one side** of $x_k$? (e.g. at the boundary node $x_1 = a$: no node on its left)

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**Idea**: one more node, $x_{k+2}$, and a combination of two expansions that cancels the second-order term

Expansion around $x_k$, evaluated at $x = x_{k+2}$:

$$f_{k+2} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k (2\Delta x) + \left.\frac{1}{2}\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k (2\Delta x)^2 + O((2\Delta x)^3)$$

$$\downarrow \quad x - x_0 = 2\Delta x\text{: } (2\Delta x)^2 = 4\Delta x^2,\ O((2\Delta x)^3) = O(\Delta x^3)$$

$$f_{k+2} = f_k + 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + 2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [3]$$

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$$f_{k+1} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \frac{1}{2}\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [1]$$

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second-order coefficients: 1/2 and 2 $\Rightarrow$ multiplying [1] by 4 makes them equal

Multiplying [1] by 4 ($4 \cdot O(\Delta x^3) = O(\Delta x^3)$):

$$4f_{k+1} = 4f_k + 4\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + 2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3)$$

Subtracting [3] member by member, term by term:

$$4f_{k+1} - f_{k+2} = (4f_k - f_k) + \left(4\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x - 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x\right) + \cancel{\left(2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 - 2\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2\right)} + O(\Delta x^3) - O(\Delta x^3)$$

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# Finite differences - one-sided forward formula

$$4f_{k+1} - f_{k+2} = 3f_k + 2\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + O(\Delta x^3)$$

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Reordering:

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k = \frac{-3f_k + 4f_{k+1} - f_{k+2}}{2\Delta x} + O(\Delta x^2)$$

**one-sided forward** formula: three nodes $x_k, x_{k+1}, x_{k+2}$, **second-order** accurate

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# Finite differences - one-sided backward formula

Same steps with the nodes $x_{k-1}$, $x_{k-2}$: $x - x_0 = -\Delta x$ and $-2\Delta x$

$$f_{k-1} = f_k - \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \frac{1}{2}\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + O(\Delta x^3) \quad [2]$$

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Subtracting $3f_k$, dividing by $-2\Delta x$, reordering:

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Numerical differentiation

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Taylor expansion

First derivative

# Second derivative

# Finite differences - second derivative: expansions

**Question**: a formula for $\mathrm{d}^2 f / \mathrm{d}x^2$ at $x_k$ from $f_{k-1}$, $f_k$, $f_{k+1}$?

**Idea**: **adding** [1] and [2], the first-derivative terms (opposite signs) cancel, the second-derivative terms add up

At the end we divide by $\Delta x^2$ (two powers lost): one more term in the expansions, error $O(\Delta x^4)$

$$f_{k+1} = f_k + \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x + \frac{1}{2}\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + \frac{1}{6}\left.\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right|_k \Delta x^3 + O(\Delta x^4) \quad [5]$$

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coefficient $1/3! = 1/6$; the third power changes sign, as $(-\Delta x)^3 = -\Delta x^3$

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# Finite differences - second derivative: adding

Adding member by member, [5] + [6], term by term:

$$f_{k+1} + f_{k-1} = 2f_k + \cancel{\left( \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x - \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_k \Delta x \right)} + 2 \cdot \frac{1}{2} \left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k \Delta x^2 + \cancel{\left( \left.\frac{1}{6}\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right|_k \Delta x^3 - \left.\frac{1}{6}\frac{\mathrm{d}^3 f}{\mathrm{d}x^3}\right|_k \Delta x^3 \right)} + O(\Delta x^4) + O(\Delta x^4)$$

$O(\Delta x^4) + O(\Delta x^4) = O(\Delta x^4)$, $2 \cdot 1/2 = 1$:

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centered finite difference for the second derivative: **second-order** accurate

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$$\frac{f_{k+1} - 2f_k + f_{k-1}}{\Delta x^2} = \left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k + \frac{O(\Delta x^4)}{\Delta x^2}$$

$$\downarrow O(\Delta x^4)/\Delta x^2 = O(\Delta x^2)$$

$$\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_k = \frac{f_{k+1} - 2f_k + f_{k-1}}{\Delta x^2} + O(\Delta x^2)$$

centered finite difference for the second derivative: **second-order** accurate

# Finite differences - truncation error

**Question**: how does the error change when the spacing is halved?

| formula | error | $\Delta x$ | $\Delta x/2$ |
|---|---|---|---|
| forward / backward | $O(\Delta x)$ | $E$ | $E/2$ |
| centered (first and second derivative) | $O(\Delta x^2)$ | $E$ | $E/4$ |
| one-sided, three nodes | $O(\Delta x^2)$ | $E$ | $E/4$ |

halving $\Delta x$ divides an $O(\Delta x^N)$ error by $2^N$

Check in the lab: errors for $\Delta x$ and $\Delta x/2$ and their ratio (step 3)

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LAB >_

# Numerical derivatives

02_numerical_derivatives

finite difference formulas vs the exact derivative; order of accuracy, truncation and round-off; plain script `numerical_derivatives.m` (sections %%)

# Lab - numerical derivatives: step 1, functions and formulas

LAB

**Lab** `02_numerical_derivatives`: finite difference formulas vs the exact derivative

1. Section `%% functions`: $f(x) = 2x^4 + 3x^3 + x^2 + 0.5x + 3$ (given) and its exact derivative as anonymous functions

```
f = @(x) 2*x.^4 + 3*x.^3 + x.^2 + 0.5*x + 3;
df_dx_e = @(x) ...
```

`.^`: element-wise power, so that `f` also accepts arrays

2. Plot $f$ and $\mathrm{d}f/\mathrm{d}x$ on $[0, 2]$ (`linspace`); two y-axes with `yyaxis left`, `yyaxis right`

3. Section `%% formulas`: one anonymous function per formula, with two inputs $x$ and $\Delta x$:

```
df_dx_FW = @(x,dx) (f(x+dx) - f(x))./dx;
```

and similarly `df_dx_BW`, `df_dx_C`

# Lab - numerical derivatives: step 2, one point

LAB

**Lab** `02_numerical_derivatives`: finite difference formulas vs the exact derivative

4. Section `%% one point`: at $x = 1$ with $\Delta x = 10^{-3}$, display the exact derivative, the three estimates and their absolute errors (`format long`)

You should get:

| | $\mathrm{d}f/\mathrm{d}x$ | error |
|---|---|---|
| exact | 19.5 | |
| FW | 19.522011002 | $2.2 \cdot 10^{-2}$ |
| BW | 19.478010998 | $2.2 \cdot 10^{-2}$ |
| C | 19.500011 | $1.1 \cdot 10^{-5}$ |

5. **Question**: why is the centered estimate not exact?

   hint: leading error term $1/6\ \mathrm{d}^3f/\mathrm{d}x^3\ \Delta x^2$, with $\mathrm{d}^3f/\mathrm{d}x^3 = 48x + 18$

# Lab - numerical derivatives: step 3, halving $\Delta x$

LAB

**Lab** `02_numerical_derivatives`: finite difference formulas vs the exact derivative

6. Section `%% halving`: errors of the three formulas at $x = 1$ for $\Delta x = 10^{-2}$ and $\Delta x/2$, and their ratio $E(\Delta x)/E(\Delta x/2)$
7. **Question**: which ratios does the theory predict?

You should get:

| | $E(10^{-2})$ | $E(0.5 \cdot 10^{-2})$ | ratio |
|---|---|---|---|
| FW | $2.21 \cdot 10^{-1}$ | $1.1 \cdot 10^{-1}$ | 2.01 |
| BW | $2.19 \cdot 10^{-1}$ | $1.1 \cdot 10^{-1}$ | 2 |
| C | $1.1 \cdot 10^{-3}$ | $2.75 \cdot 10^{-4}$ | 4 |

$\approx 2 = 2^1$ (first order), $\approx 4 = 2^2$ (second order)

# Lab - numerical derivatives: step 4, error versus $\Delta x$

LAB

**Lab** `02_numerical_derivatives`: finite difference formulas vs the exact derivative

8. Section `%% error versus dx`: $\Delta x$ from $10^{-1}$ to $10^{-16}$, table of the exact value and of the three estimates

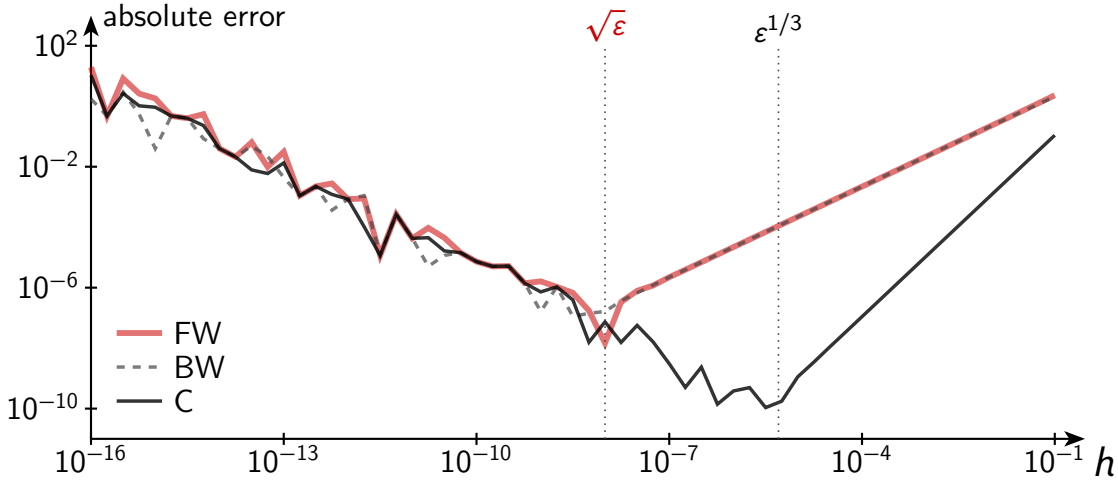
```
dx = logspace(-1,-16,16);
T = table(..., 'VariableNames',["exact","dx","FW","BW","C"]);
```

9. Absolute errors for 250 values of $\Delta x$, logarithmic plot (`loglog`)

10. **Questions**: slope of each curve for large $\Delta x$? Where is the minimum? Why does the error grow again as $\Delta x \to 0$?

# Lab - numerical derivatives: step 4, result

**Lab** Absolute error at $x = 1$ versus $h = \Delta x$ (computed in double precision)




Large $h$: slope 1 (FW, BW), slope 2 (C): **truncation**. Small $h$: slope $-1$, **round-off**; minima of the order of $\sqrt{\varepsilon}$ and $\varepsilon^{1/3}$ (noisy: round-off)

# Lab - numerical derivatives: step 5, second derivative

LAB

**Lab** `02_numerical_derivatives`: finite difference formulas vs the exact derivative

11. Section `%% second derivative`: anonymous functions for the centered formula and for the exact $\mathrm{d}^2 f/\mathrm{d}x^2 = 24x^2 + 18x + 2$; errors at $x = 1$ for $\Delta x = 10^{-2}$ and $\Delta x/2$, their ratio; `loglog` of the error for `dx = logspace(-1,-8,200)`

```
d2f_dx2_C = @(x,dx) (f(x+dx) - 2*f(x) + f(x-dx))./dx.^2;
```

You should get: $E = 4 \cdot 10^{-4}$ and $10 \cdot 10^{-5}$, ratio 4

12. **Questions**: why the ratio 4? Where is the minimum of the error now, and why?
    hint: round-off of the numerator divided by $\Delta x^2$

# Finite differences - truncation and round-off

**Question**: why can't the error be made arbitrarily small by reducing $\Delta x = h$?

- **Truncation** error: finite number of Taylor terms (derivative $\rightarrow$ algebraic expression)
- **Round-off** error: finite number of digits

Forward formula in floating point: $f_{k+1}$ and $f_k$ carry relative errors up to $\varepsilon$

$$\frac{\mathrm{fl}(f_{k+1}) - \mathrm{fl}(f_k)}{h} = \frac{f_{k+1} - f_k}{h} + O(\varepsilon \left|f_k\right| / h)$$

$f_{k+1} \approx f_k$ for small $h$: subtraction of close numbers (cancellation)

Total error: truncation decreases with $h$, round-off increases as $h \rightarrow 0$

$$|E(h)| \approx \underbrace{Mh}_{\text{truncation}} + \underbrace{\varepsilon \left|f\right| / h}_{\text{round-off}}$$

Minimum where the two terms are comparable: $h \approx \sqrt{\varepsilon} \approx 10^{-8}$ for the forward formula; $h \approx \varepsilon^{1/3} \approx 10^{-5}$ for the centered one (truncation $Mh^2$)

higher order: smaller minimum error, reached with a larger $h$

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# Numerical differentiation - summary

- Derivatives are approximated by combinations of **nodal values**, obtained from Taylor expansions
- Forward and backward: $O(\Delta x)$; centered (first and second derivative): $O(\Delta x^2)$
- Truncation error decreases with $\Delta x$; round-off error grows as $\Delta x \to 0$
- Next step: replace the derivatives of a **differential equation** with these formulas $\to$ finite difference method

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# Advanced - from derivative to Jacobian

ADVANCED



Consider $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$, e.g. the residual $\boldsymbol{F}(\boldsymbol{u})$ of a discretised **nonlinear** problem, $\boldsymbol{u}$: nodal values

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**Jacobian matrix** $(m \times n)$: derivative of each component with respect to each variable

$$J_{ij} = \frac{\partial f_i}{\partial x_j}, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

$n = m = 1$: $J = \mathrm{d}f / \mathrm{d}x$

Taylor expansion to first order: $\boldsymbol{f}(\boldsymbol{x} + \Delta\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x}) + J\Delta\boldsymbol{x} + O(\|\Delta\boldsymbol{x}\|^2)$

Newton method for $\boldsymbol{F}(\boldsymbol{u}) = \boldsymbol{0}$: at each iteration

$$J\left(\boldsymbol{u}^{(k)}\right) \delta\boldsymbol{u} = -\boldsymbol{F}\left(\boldsymbol{u}^{(k)}\right), \quad \boldsymbol{u}^{(k+1)} = \boldsymbol{u}^{(k)} + \delta\boldsymbol{u}$$

e.g. saturable ferromagnetic materials ($\mu$ depends on $|\boldsymbol{H}|$), temperature-dependent $\sigma$; quadratic convergence requires an accurate $J$

Also: **sensitivities** of an output (force, inductance) with respect to design parameters

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# Advanced - Jacobian by finite differences

ADVANCED



Column $j$ of $J$: forward formula along the direction $\boldsymbol{e}_j$ (unit vector of variable $x_j$)

$$J\boldsymbol{e}_j \approx \frac{\boldsymbol{f}\left(\boldsymbol{x} + h\boldsymbol{e}_j\right) - \boldsymbol{f}\left(\boldsymbol{x}\right)}{h}, \quad j = 1, \ldots, n$$

Cost: $n + 1$ evaluations of $\boldsymbol{f}$ ($2n$ with the centered formula)

$n$: number of unknowns, $10^4$–$10^6$ in a field model

Accuracy: same trade-off as in the scalar case; best relative error $\approx \sqrt{\varepsilon}$ (forward), $\approx \varepsilon^{2/3}$ (centered)

| format | $\varepsilon$ | forward, $\sqrt{\varepsilon}$ | centered, $\varepsilon^{2/3}$ |
|---|---|---|---|
| double | $1.1 \cdot 10^{-16}$ | $1.1 \cdot 10^{-8}$ | $2.3 \cdot 10^{-11}$ |
| single | $6 \cdot 10^{-8}$ | $2.4 \cdot 10^{-4}$ | $1.5 \cdot 10^{-5}$ |
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Finite difference Jacobian: $n + 1$ evaluations, about **half** the digits, $h$ to be tuned

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# Advanced - automatic differentiation

ADVANCED



A program computing $f$ is a sequence of elementary operations ($+$, $\times$, sin, exp, ...) with known derivatives

**Automatic differentiation** (AD): chain rule applied to each operation, on numbers

Forward mode: each variable carries the pair (value, derivative) $(a, a')$

$$(a, a') + (b, b') = (a + b, a' + b'), \quad (a, a') \cdot (b, b') = (ab, a'b + ab'), \quad \sin(a, a') = (\sin a, a' \cos a)$$

Example, $f(x) = x \sin x$: starting from $(x_0, 1)$

$$(x_0, 1) \cdot (\sin x_0, \cos x_0) = (x_0 \sin x_0, \ \sin x_0 + x_0 \cos x_0)$$

Lab function at $x = 1$, with pairs: $f = 9.5$, $\mathrm{d}f/\mathrm{d}x = 19.5$ (exact: 19.5)

computed live with (value, derivative) pairs in double precision

> No $h$, no truncation error: derivative exact up to the round-off of the elementary operations

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**Automatic differentiation** (AD): chain rule applied to each operation, on numbers

Forward mode: each variable carries the pair (value, derivative) $(a, a')$

$$(a, a') + (b, b') = (a + b, a' + b'), \quad (a, a') \cdot (b, b') = (ab, a'b + ab'), \quad \sin(a, a') = (\sin a, a' \cos a)$$

Example, $f(x) = x \sin x$: starting from $(x_0, 1)$

$$(x_0, 1) \cdot (\sin x_0, \cos x_0) = (x_0 \sin x_0, \quad \sin x_0 + x_0 \cos x_0)$$

Lab function at $x = 1$, with pairs: $f = 9.5$, $\mathrm{d}f/\mathrm{d}x = 19.5$ (exact: 19.5)

computed live with (value, derivative) pairs in double precision

> No $h$, no truncation error: derivative exact up to the round-off of the elementary operations

# Advanced - automatic differentiation: cost

ADVANCED



Jacobian of $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$, one pass: a small multiple of the cost of $\boldsymbol{f}$

Forward mode: one pass per **input** direction $\boldsymbol{e}_j \to J$ column by column, $n$ passes

Reverse mode (adjoint): one pass per **output** $\to J$ row by row, $m$ passes

scalar output (energy, force, cost function): gradient with respect to all parameters in one pass, independent of $n$; intermediate values stored in memory

Sparse $J$ (FDM, FEM: each equation involves few nodes): columns without common nonzero rows computed together (colouring)

number of passes $\approx$ nonzeros per row, instead of $n$; also valid for finite differences

| | accuracy | passes for $J$ | parameter $h$ |
|---|---|---|---|
| finite differences | $\approx \sqrt{\varepsilon}$ | $n+1$ | yes |
| AD, forward | $\approx \varepsilon$ | $n$ | no |
| AD, reverse | $\approx \varepsilon$ | $m$ | no |

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# Advanced - AD tools and thesis activity

ADVANCED



Implementations:

- operator overloading (C++): autodiff, CoDiPack, CppAD, Adept, Sacado (Trilinos)
- source or compiler transformation: Tapenade, Enzyme (LLVM)
- MATLAB: `dlgradient` (Deep Learning Toolbox), automatic derivatives in the Optimization Toolbox

**Thesis activity** Jacobians by AD in C++ for differential-algebraic problems of non-thermal plasmas (code Merlino2d):

- transport of the plasma species coupled with Poisson's equation: differential-algebraic system
- implicit time integration: Newton method, Jacobian at each iteration
- Jacobian from AD versus finite differences: accuracy, Newton convergence, computational cost

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